D. Divjak. *Frequency in language: Memory, attention and learning.* Cambridge: Cambridge University Press, 2019. ISBN 9781107085756 xiv+327 pp. https://doi.org/10.1017/9781316084410

Reviewed by Tsy Yih (Zhejiang University)

Modern linguistics is often said to be a part of the cognitive sciences (Isac & Reiss, 2013). However, the classical issues in the latter field, such as memory, attention, reasoning or problem solving, have rarely been found in (cognitive) linguistic textbooks (e.g., Evans, 2019). In this regard, *Frequency in language: Memory, attention and learning* (henceforth *FiL*), by Dagmar Divjak, is such a bold move, as it attempts to bridge the gap between the fields of cognitive linguistics and cognitive science in general. *FiL* primarily addresses a key question: How is the Cognitive Commitment manifested in linguistics? In other words, it offers an in-depth exposition of how domain-general cognitive abilities are employed to process language in the usage-based approach to linguistic studies.

The whole monograph is divided into four parts. While all parts are untitled, the gist can be easily grasped as readers go through them, viz. frequency, memory, attention and learning. It is also what the title and subtitle of the book indicate. In this review, I would firstly give a brief summary of each chapter as follows.

Chapter 1 sets out from the dispute between naturists and nurturists over the role that frequency plays in language development. A brief history of frequency studies in lexical statistics, psycholinguistics and quantitative linguistics are then reviewed. Chapter 2 focuses on the role of frequency in several usage-based approaches and the way to measure occurrence and co-occurrence in the existing corpus linguistic literature. Chapter 3 introduces probabilistic grammar that relies on the data of frequency, thereby linking linguistics to information theory. In Chapter 4, the author points out that researchers try to understand memory through a variety of metaphors, which may help or hinder our understanding of its very nature. Meanwhile, the multi-store model of memory (sensory, working and long-term) and three types of memory processes (encoding, storage and retrieval) are also presented. The author then discusses the concept of 'entrenchment' put forth by Ronald Langacker in linguistics and explores in Chapter 5 how it can be related to memory in general. Chapter 6 reviews the brain's attention mechanism, while Chapter 7 is devoted to its linguistic counterpart, i.e., salience. According to the author, salience has at least four usages in the linguistic literature (p. 195) and has been employed as the explanans in a number of case stud-

https://doi.org/10.1075/rcl.00097.yih
Review of Cognitive Linguistics 19:2 (2021), pp. 596–601. ISSN 1877-9751 | E-ISSN 1877-976X
© John Benjamins Publishing Company

ies. Chapters 8 and 9 address the issue of prediction and learning, respectively. In Chapter 8, the author offers an exposition of the well-known three-level model by David Marr, which was put forward in the field of vision and has been widely accepted as a foundation of cognitive science. The last concrete chapter pays special attention to the existent learning theories in psychology and its applications in linguistics after their divorce six decades ago (p. 234).

After summarizing the entire book, here are some points I would like to delve into. First and foremost, speaking of frequency, it is inevitable to mention the field of quantitative linguistics, since it plays a prominent part in this subdiscipline. The author also assigns a separate section (Section 1.4) to the theme, regarding Kingsley Zipf, G. Udny Yule and Gustave Herdan as "three of the first and most well-known researchers in the area of word frequency distributions from the Anglo-Saxon tradition" (p. 30). However, Divjak does not mention much about the later development of this subdiscipline, especially the mainstream Altmann school (Köhler, 2012, p. 7; Liu, 2017, Section 1.4). The Altmann school of quantitative linguistics was originally and primarily thriving in Central Europe, and later attracted scholars' attention from other parts of the world. Apart from Gabriel Altmann, after whom the school is named, there are many other well-known names, including Reinhard Köhler and Karl-Heinz Best from Germany, Peter Grzybek from Austria, Ioan-Iovitz Popescu from Italy, Ramon Ferrer-i-Cancho from Spain, Haitao Liu from China, Haruko Sanada from Japan, etc. Most of them have educational backgrounds in science and engineering. Traditionally, quantitative linguistics is also called text studies which seldom relate their results to human cognition. Nonetheless, more recently, quantitative linguists have stepped out of their small circle and started to publish in a number of international journals of cognitive science and interdisciplinary studies. For instance, Liu (2008) found that the minimalization of mean dependency distance is a demonstration of how frequency helps shape linguistic structures that are constrained by the limit of short-term memory (see Liu et al., 2017 for a detailed overview). Such attempts emphasize the role that language users play in the process. Apart from quantitative linguistics, another notable trend lies in the neo-Firthian corpus linguistics. M.A.K. Halliday (1993), following his teacher, proposed a usage-based point of view which was further termed by Schmid (2000) the 'From-Corpus-to-Cognition Principle'. Stefanowitsch and Flach (2017) later called it the 'corpus-as-input view', as opposed to the 'corpus-as-output view' (Schmid, 2020, p. 217).[1] The idea behind this coincides with Liu and other scholars' statement that language is a human-driven complex adaptive system. In summary, no matter which school researchers come from or what educational backgrounds they have, they

1. I would like to thank the anonymous referee for pointing that out.

are reaching converging agreement on universal cognitive constraints imposed on language processing and on how they shape linguistic structures. This is also one of the major concerns of cognitive linguistics.

Second, there is a reiterated point worth noticing which also constitutes the motivation of this book. That is, cognitive linguistics, in the past decades since its establishment, has given a much higher priority to categorization and metaphor/metonymy than to other issues in cognitive science. Therefore, cognitive linguists, on the one hand, are encouraged to pay more attention to other cognitive abilities. Apart from memory, attention and learning being fully discussed in the book, a few more topics could have been added. For instance, while perception is a well-studied area in auditory phonetics, the author has given little space to it (although she did mention it in her introduction of a handbook mentioned beneath). Higher-level cognition processes such as decision making and reasoning are also beyond the scope of this book, which remain open for future research. On the other hand, it would be interesting to look backward and see how cognitive linguistics can contribute to the broader field of cognitive science. Taking categorization as an example, besides the formal/computational models proposed in cognitive science (cf. Pothos & Wills, 2011), linguistic approaches to this issue have offered abundant empirical case studies. Combined together, they are bound to deepen our understanding of this cognitive process.

Third, and related to the second point, it has long been criticized that "formal linguistics is not formal" (with primary reference to the generative trend after the 1970s, c.f. Müller, 2018) and that "cognitive linguistics is not cognitive" in the sense of cognitive science. However, we are relieved to see some scholars still endeavoring to pull both fields back on the right track, such as Edward Stabler, who has invariably devoted himself to the formalization of generative grammar. As for the latter case, we are lucky to have Dagmar Divjak, who is the editor-in-chief of *Cognitive Linguistics*, one of the flagship journals of the area. In 2015, Divjak briefly mentioned those cognitive abilities in the introductory chapter of the *Handbook of Cognitive Linguistics* (co-authored with Ewa Dąbrowska). Her established status will definitely make this self-reflection within the cognitive linguists' community carry more weight and draw more attention. Methodologically, cognitive linguistics has surpassed the generative school since, except for a few persistent scholars, most generative grammarians barely use strict formalization in their approaches, while cognitive linguists have been employing more advanced techniques such as machine learning and have seen an apparent quantitative turn (Janda, 2013).

However, this book, like every other book, has some unavoidable but minor limitations with regard to the contents. First, Chapter 2 devotes some space to "Bybee's exemplar theory", while there are more advocates of the exemplar theory

other than Bybee. The debate between the prototype and the exemplar's view on categorization has a long history in psychology, and converging evidence shows that both strategies are involved in human mind processes (Vanpaemel & Storms, 2008; Smith, 2014). I believe the author definitely knows the distinction (cf. Divjak & Arppe, 2013) but simply fails to demonstrate it to readers here. Second, besides 'salience', a few other linguistic notions related to attention are left undiscussed in the book. One such notion, though vague as well, is 'prominence'. Others include the 'theme/rheme' distinction which dates back to the Prague school. In one word, the review part of information structure in *Fil* could be expected to be more comprehensive given the vast literature in various schools and traditions.

Last but not least, I have a few comments on the structure of the book. Most parts of the book are reader-friendly and easy to grasp. For instance, Parts II and III each contain two chapters which go in parallel. The first chapter in each part introduces the cognitive concept in general, and the second one links it to the corresponding notion originating in linguistics. This is also the case for Chapter 9 which deals with the learning theory. However, for Chapters 1 and 2, it is easy to get lost, since the relations between sections seem to be loose and hard to follow. For instance, I think it would be better to combine the last three sections in Chapter 1 with Section 2.1 into a new chapter, since, unlike Section 1.1, they all review pro-frequency research traditions. In addition, Section 2.2 focuses on association/collocation, which is simply an indirect measure calculated on the basis of the directly observed (co-)occurrence counts in corpora or texts. This is similar to the case of Section 3.3 where other indirect measures such as surprisal and entropy are introduced. All the above-mentioned measures reflect some of our intuitions on frequency-related linguistic properties. Hence, Section 2.2 might better be placed in Chapter 3 under the title of "more than frequencies". Therefore, an ideal arrangement might be as follows: Section 1.1 could first stand as a separate introductory chapter. Then Sections 1.2 to 1.4 and Section 2.1 could be combined into a whole new chapter dedicated to the literature overview of frequency studies in different schools and subdisciplines. Finally, Section 2.2 could be merged into Chapter 3, which concentrates on frequency-based indirect measures. By doing so, the logic of the book would, from my point of view, be clearer to readers.

Overall, *FiL* refers extensively to psychology, as well as to neuroscience, corpus linguistics, quantitative linguistics, information theory, computer vision, etc. This makes it an informative and multidisciplinary volume, which provides linguists who do not have backgrounds in these fields with the knowledge they need. Moreover, the author insightfully matches several pairs of notions originating respectively in cognitive linguistics and cognitive science (e.g. "entrenchment" vs. "memory", "salience" vs. "attention"), thereby offering a new perspective to the

reader. Although the structure of the book can be more well-organized and the content more balanced, all these minor problems will not overshadow its merits.

References

Divjak, D., & Arppe, A. (2013). Extracting prototypes from exemplars: What can corpus data tell us about concept representation? *Cognitive Linguistics 24*(2), 221–274. https://doi.org/10.1515/cog-2013-0008

Evans, V. (2019). *Cognitive Linguistics A complete guide (2nd ed.)*. Edinburgh: Edinburgh University Press.

Halliday, M.A.K. (1993). Quantitative studies and probabilities in grammar. In M. Hoey (Ed.), *Data, description, discourse: Papers on the English language in honour of John McH. Sinclair* (pp. 1–25). London: HarperCollins.

Isac, D., & Reiss, C. (2013). *I-Language: An introduction to linguistics as cognitive science (2nd ed.)*. Oxford: Oxford University Press.

Janda, L.A. (2013). Quantitative methods in Cognitive Linguistics: An introduction. In L.A. Janda (ed.), *Cognitive Linguistics: The quantitative turn* (pp. 1–32). Berlin & New York: Mouton de Gruyter. https://doi.org/10.1515/9783110335255.1

Köhler, R. (2012). *Quantitative syntax analysis*. Berlin & New York: Mouton de Gruyter. https://doi.org/10.1515/9783110272925

Liu, H. (2008). Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science 9*(2), 159–191. https://doi.org/10.17791/jcs.2008.9.2.159

Liu, H. (2017). *An introduction to quantitative linguistics*. Beijing: The Commercial Press.

Liu, H., Xu, C., & Liang, J. (2017). Dependency distance: A new perspective on syntactic patterns innatural languages. *Physics of Life Reviews*, *21*, 171–193. https://doi.org/10.1016/j.plrev.2017.03.002

Müller, S. (2018). *Grammatical theory: From transformational grammar to constraint-based approaches (2nd ed.)*. Berlin: Language Science Press.

Pothos, E.M., & Wills, A.J., (Eds.). (2011). *Formal approaches in categorization*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511921322

Schmid, H.-J. (2000). *English abstract nouns as conceptual shells: From corpus to cognition*. Berlin & New York: Mouton de Gruyter. https://doi.org/10.1515/9783110808704

Schmid, H.-J. (2020). *The dynamics of the linguistic system: Usage, conventionalization, and entrenchment*. Oxford: Oxford University Press. https://doi.org/10.1093/oso/9780198814771.001.0001

Smith, J.D. (2014). Prototypes, exemplars, and the natural history of categorization. *Psychonomic Bulletin & Review*, *21*(2), 312–331. https://doi.org/10.3758/s13423-013-0506-0

Stefanowitsch, A., & Flach, S. (2017). The corpus-based perspective on entrenchment. In H.-J. Schmid (Ed.), *Entrenchment and the psychology of language learning: How we reorganize and adapt linguistic knowledge* (pp. 101–127). Berlin & New York: Mouton de Gruyter. https://doi.org/10.1037/15969-006

Vanpaemel, W., & Storms, G. (2008). In search of abstraction: The varying abstraction model of categorization. *Psychonomic Bulletin & Review*, *15*(4), 732–749. https://doi.org/10.3758/PBR.15.4.732

Address for correspondence

Tsy Yih
School of International Studies
Zhejiang University
No. 866, Yuhangtang Road
Xihu District, Hangzhou 310058
P.R. China
yezi_leafy@hotmail.com

Biographical notes

Tsy Yih is a PhD candidate at the School of International Studies of Zhejiang University, China. Tsy Yih is the transliteration of the name of the reviewer in his mother tongue, Shanghai Wu Chinese. He is also known as ZI YE in Mandarin pinyin. His current research projects focus on quantitative linguistics and the usage-based approach to functional typology.

Publication history

Date received: 23 October 2019
Date accepted: 28 October 2020